

Prescribed-Coefficient Greedy Convergence for Countable Orthogonal-Block Dictionaries

A substantial partial answer to the question in arXiv:2209.03091

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Abstract

Oliaro–Tomatis–Valiullin–Valiullin ask for a characterization of the symmetric Hilbert-space dictionaries for which every exact greedy expansion converges for every positive coefficient sequence $c_n \rightarrow 0$ with $\sum_n c_n = \infty$. We prove convergence for a new countable orthogonal-block class. Each block may be infinite-dimensional and carry an arbitrary symmetric dictionary, provided all block dictionaries have a common positive norming constant and attain their local maxima. In particular, an orthonormal basis may be augmented by arbitrary closed sets of atoms in infinitely many pairwise disjoint finite coordinate blocks of uniformly bounded size. The source allowed extra atoms only in one fixed finite block.

Status. substantial partial result; likely valid. The full characterization remains open.

1 Source question and definitions

Let H be a real Hilbert space. A symmetric dictionary $\mathcal{D}$ is a symmetric set of unit vectors with dense linear span. Put

$$\rho_{\mathcal{D}}(x) = \sup_{g \in \mathcal{D}} \langle x, g \rangle = \sup_{g \in \mathcal{D}} |\langle x, g \rangle|.$$

For positive prescribed coefficients (c_n) , the exact greedy algorithm starts with $r_0 = f$ and, while $r_{n-1} \neq 0$, chooses $g_n \in \mathcal{D}$ satisfying

$$\langle r_{n-1}, g_n \rangle = \rho_{\mathcal{D}}(r_{n-1}), \quad r_n = r_{n-1} - c_n g_n.$$

Convergence means $r_n \rightarrow 0$ for every initial f and every possible exact maximizer sequence.

The source proves this when $\mathcal{D}$ is a symmetrized orthonormal basis, and when arbitrary additional atoms are supported on one fixed finite subset of that basis. Its concluding question asks for a characterization of all dictionaries having convergence under

$$c_n > 0, \quad c_n \longrightarrow 0, \quad \sum_{n=1}^{\infty} c_n = \infty. \tag{1}$$

5 Conclusions

In this paper we have extended the convergence result of the greedy algorithm with prescribed coefficients known for finite dimensional Hilbert spaces, and we have proved that such a result, in the case of a weakening parameter $t = 1$, holds also in the case of infinite dimensional spaces when the dictionary is the symmetrized of an orthonormal basis, to which linear combinations of elements from a finite subset of the basis itself can eventually be joined. These are sufficient conditions on the dictionary to have convergence of the algorithm; the question to characterize the dictionaries for which the greedy algorithm converges under the considered hypotheses on the coefficients remains open.

2 Proof intuition

Write the Hilbert space as an orthogonal sum of blocks. Greedy selection then compares one scalar quantity from each block: its local dictionary support. For the initial vector, order those local supports as $s_1 \geq s_2 \geq \dots$. Before the first time the global greedy support falls below s_N , no block after N can have been touched. At that stopping time, the uniform local norming condition bounds each of the first N block norms by s_N/α , while the remaining tail is still the initial tail. Since $(s_j) \in \ell_2$ and is decreasing, $Ns_N^2 \rightarrow 0$, forcing a subsequence of remainders to tend to zero.

To upgrade subsequential decay to full convergence, we use an exact-greedy Lyapunov quantity that seems particularly well suited to this question:

$$E_n = \|r_n\|^2 - \rho_{\mathcal{D}}(r_n)^2.$$

Symmetry and exact maximization make (E_n) nonincreasing. Once $\liminf \|r_n\| = 0$, this defect tends to zero, so any remainder bounded away from zero is almost parallel to a maximizing atom and its norm must decrease.

3 A general exact-greedy lemma

Lemma 1 (Subsequential decay forces convergence). *Let $\mathcal{D}$ be a symmetric dictionary for which the exact maximum is attained at every remainder. Suppose $c_n \rightarrow 0$. For any exact greedy trajectory,*

$$\liminf_{n \rightarrow \infty} \|r_n\| = 0 \implies \lim_{n \rightarrow \infty} \|r_n\| = 0.$$

Proof. Set $a_n = \rho_{\mathcal{D}}(r_n)$ and $E_n = \|r_n\|^2 - a_n^2 \geq 0$. If g_{n+1} is chosen at step $n+1$, then $\langle r_n, g_{n+1} \rangle = a_n$. Symmetry gives

$$a_{n+1} \geq |\langle r_n - c_{n+1}g_{n+1}, g_{n+1} \rangle| = |a_n - c_{n+1}|.$$

Consequently,

$$\begin{aligned} E_{n+1} &\leq \|r_n\|^2 - 2c_{n+1}a_n + c_{n+1}^2 - (a_n - c_{n+1})^2 \\ &= E_n. \end{aligned}$$

Thus E_n decreases. The $\liminf$ hypothesis and $0 \leq E_n \leq \|r_n\|^2$ force $E_n \rightarrow 0$.

Fix $\varepsilon > 0$. On a sufficiently late tail, $E_n \leq \varepsilon^2/16$ and $c_{n+1} \leq \varepsilon/4$. Whenever $\|r_n\| \geq \varepsilon$ on that tail,

$$a_n \geq \sqrt{\|r_n\|^2 - E_n} \geq \frac{\sqrt{15}}{4}\varepsilon,$$

and hence

$$\|r_{n+1}\|^2 - \|r_n\|^2 = -2c_{n+1}a_n + c_{n+1}^2 < 0.$$

Choose a still later index k with $\|r_k\| < \varepsilon$. If a subsequent remainder is below ε , the next has norm less than $\varepsilon + c_{n+1} \leq 5\varepsilon/4$; while its norm is at least ε , it decreases. Therefore $\sup_{n \geq k} \|r_n\| \leq 5\varepsilon/4$. Since ε is arbitrary, $r_n \rightarrow 0$. $\square$

4 Countable orthogonal-block theorem

Theorem 2. *Let*

$$H = \bigoplus_{j=1}^{\infty} H_j$$

be an orthogonal Hilbert sum. For each j , let $\mathcal{D}_j$ be a symmetric unit dictionary in H_j such that every local support maximum

$$\rho_j(x) = \max_{g \in \mathcal{D}_j} \langle x, g \rangle$$

is attained. Suppose there is one $\alpha > 0$ such that

$$\rho_j(x) \geq \alpha \|x\| \quad (x \in H_j, j \geq 1). \quad (2)$$

Embed the $\mathcal{D}_j$ into H and put $\mathcal{D} = \bigcup_j \mathcal{D}_j$. Then for every coefficient sequence satisfying (1), every exact greedy expansion with respect to $\mathcal{D}$ converges to its initial vector.

Proof. For $x = (x_j) \in H$, the numbers $\rho_j(x_j)$ tend to zero, since they are at most $\|x_j\|$. Thus the global supremum is a maximum whenever $x \neq 0$, and $\mathcal{D}$ is an admissible symmetric dictionary.

Let (r_n) be a greedy trajectory and write $a_n = \rho_{\mathcal{D}}(r_n)$. The energy identity is

$$\|r_{n+1}\|^2 = \|r_n\|^2 - 2c_{n+1}a_n + c_{n+1}^2. \quad (3)$$

We first have $\liminf_n a_n = 0$. Otherwise $a_n \geq \delta > 0$ eventually; once $c_{n+1} \leq \delta$, equation (3) gives $\|r_{n+1}\|^2 \leq \|r_n\|^2 - \delta c_{n+1}$. The divergent sum in (1) then forces a negative squared norm.

Write the initial vector as $f = (f_j)$ and relabel the blocks so that

$$s_j := \rho_j(f_j), \quad s_1 \geq s_2 \geq \dots \geq 0.$$

We have $\sum_j s_j^2 \leq \sum_j \|f_j\|^2 < \infty$. If only finitely many s_j are nonzero, the trajectory lies in a finite orthogonal sum on which

$$\rho_{\mathcal{D}}(x) = \max_j \rho_j(x_j) \geq \frac{\alpha}{\sqrt{N}} \|x\|,$$

and the same energy argument as above gives convergence. We may therefore assume all $s_N > 0$.

For each N , let m_N be the first index with $a_{m_N} \leq s_N$; it exists because $\liminf a_n = 0$. Every block $j > N$ is untouched before this time. Indeed, while it is untouched its local support is $s_j \leq s_N$, whereas at every earlier step the selected global support is strictly larger than s_N . At time m_N , condition (2) therefore gives

$$\|(r_{m_N})_j\| \leq \frac{s_N}{\alpha} \quad (j \leq N), \quad (r_{m_N})_j = f_j \quad (j > N).$$

Hence

$$\|r_{m_N}\|^2 \leq \frac{Ns_N^2}{\alpha^2} + \sum_{j>N} \|f_j\|^2. \quad (4)$$

Since (s_j) is decreasing and square summable,

$$\frac{N}{2} s_N^2 \leq \sum_{j=\lceil N/2 \rceil}^N s_j^2 \longrightarrow 0.$$

The other term in (4) is the tail of the square-summable block norms. Thus $\|r_{m_N}\| \rightarrow 0$. Lemma 1 completes the proof. $\square$

5 Infinitely many finite perturbation blocks

Corollary 3. *Let $(e_i)_{i \in I}$ be an orthonormal basis and partition I into finite sets I_j satisfying $|I_j| \leq M < \infty$. In each $H_j = \text{span}\{e_i : i \in I_j\}$, let Y_j be any closed symmetric set of unit vectors. Then*

$$\mathcal{D} = \{\pm e_i : i \in I\} \cup \bigcup_{j=1}^{\infty} Y_j$$

has the convergence property of Theorem 2.

Proof. For the local dictionary consisting of the signed coordinate vectors and Y_j , compactness gives attainment, and

$$\rho_j(x) \geq \max_{i \in I_j} |\langle x, e_i \rangle| \geq \frac{1}{\sqrt{M}} \|x\|.$$

Apply Theorem 2 with $\alpha = M^{-1/2}$. □

When infinitely many Y_j are nonempty, their union is not supported on one fixed finite subset of the basis. Corollary 3 is therefore a strict extension of the source’s Corollary 4.3.

6 Scope, verification, and novelty status

The theorem is limited to exact greediness $t = 1$, orthogonal component dictionaries, and a uniform positive local norming constant. It does not characterize nonorthogonal dictionaries such as general Riesz bases, and it does not settle what happens when the best local constants tend to zero. Those two upgrade routes were examined; Gram-matrix interactions destroy the untouched-tail argument, while no rigorous counterexample was obtained in the nonuniform block case.

The proof was audited through the two independent monotonic mechanisms $\liminf a_n = 0$ from (3) and monotonicity of $\|r_n\|^2 - a_n^2$ in Lemma 1. A bounded search of the exact title, arXiv id, characterization wording, and citing literature found the 2025 journal version of the source, which retains the open question, but no later solution or countable-block theorem. Human mathematical and literature review is recommended.

References

- [1] A. Oliaro, L. Tomatis, A. R. Valiullin, and A. R. Valiullin, *Greedy expansions with prescribed coefficients in Hilbert spaces for special classes of dictionaries*, Constructive Approximation 62 (2025), 243–259; arXiv:2209.03091. See Corollary 4.3 and Section 5.